

be performed in lesser space. In order to achieve this, power distribution will be simplified from current three-stage structure (rectifier → inverter → single-phase power factor correction) to two-stage structure (three-phase power factor correction front-end → high voltage direct current bus).

In the market

The fact that most big players in IC manufacturing have their signature power reduction technologies itself proves how seriously the industry takes power management issues. eXtreme low power (XLP) by Microchip, fin-shaped field effect transistor (FinFET) by Taiwan Semiconductor Manufacturing Company (TSMC), fully-depleted silicon-on-insulator (FD-SOI) by STMicroelectronics and picoPower by Atmel are some examples.

We also have digitally-enhanced power analogue products in the market that offer a combination of digital configuration and analogue control, providing fast-responding, high-efficiency power conversions with firmware configurations. These also offer high-power-density solutions, excellent accuracy and digital communication interfaces.

Some of the power management components manufacturers and distributors provide support to customers for selecting the right technologies and customisation standard off-the-shelf solutions for the given application(s), while reducing design time and cost.

Keeping up with consumer needs

It has been observed that the push for better efficiency and lower power is greater in the consumer electronics segment. The boom of Internet of things (IoT) technology has reinforced this drive. This increases the value of end products in many consumer markets, with longer run times, smaller and lighter portable products, and increased efficiencies in white goods, computing, automotive and industrial applications. Users want laptops,

Continuously evolving

Evolutions in power management are being driven by factors like compliance and regulation changes, consumer expectations and cost, to name a few. New technologies are expected to trim power wastage, wherever possible.

"There are three possible approaches for power management—using right components and chips, using appropriate technology through a mix of circuit and chips and deploying intelligent design of hardware and software," commented T. Anand, whose firm specialises in the development of innovative products and prototyping. While component manufacturers are focussing on making smarter chips with lower power consumption and better functionalities, designers are concentrating on reducing resistance. Latest products in wearable electronics and consumer segment opt for intelligent software deployment and reducing dependency on hardware or chip designs. This move not only gives full control to the designers for efficient power management, but also makes design more flexible and adaptable to a certain extent.

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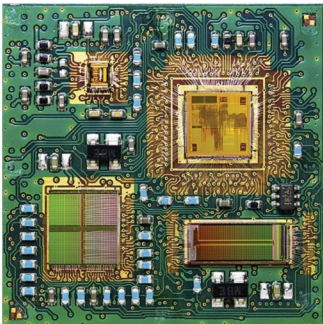


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phones, tablets and wearable devices to run continuously for longer time without recharging. Regulatory requirements for home and office appliances are becoming more stringent in many countries around the world. Automotive and infotainment industries are adding latest electronic features to vehicles and are trying to increase fuel economy at the same time. With all these developments taking place, research and development in power management segment is being given utmost importance and we can expect continu-



Inside view of a multi-chip module (MCM)
(Courtesy: <http://developer.axis.com>)

ous improvements in this area. ●

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